

EDUCATION

- **Christian-Albrechts-Universität zu Kiel** Kiel, Germany
Ph.D. in Computer Science Nov 2020 – Present
 - **Topic:** Underwater Refractive Camera Calibration and 3D Scene Reconstruction
 - **Advisor & Reference:** Prof. Dr. -Ing. Kevin Köser
- **Chongqing University** Chongqing, China
Master of Science in Surveying and Mapping Engineering (Geo-informatics) Sep 2017 – June 2020
- **Chongqing University** Chongqing, China
Bachelor of Science in Surveying and Mapping Engineering (Geo-informatics) Sep 2013 – June 2017

EXPERIENCE

- **Christian-Albrechts-Universität zu Kiel** Kiel, Germany
Computer Vision Research Assistant at Marine Data Science Group (MDS) July 2023 – Present
 - **Open-Source C++ Underwater Mapping Software:** Integrate the refractive camera models and refractive SfM into the popular C++ based open-source 3D reconstruction software COLMAP. (project link: [gitlab:colmap-underwater](https://github.com/colmap/underwater))
 - **Underwater Neural Radiance Field:** Learning a medium- and light-independent scene representation by modeling the camera-attached co-moving light source with spatial- and surface normal-dependent neural network (paper: [1]).
 - **Tiny- and Semi-transparent Object Reconstruction:** Learning a volumetric representation for tiny- and semi-transparent objects using differentiable rendering and neural radiance field (paper: [8]).
 - **Teaching Assistant - Probabilistic Robotics (Winter 2023):** Responsible for the tutorial sessions and the final exam.
- **Helmholtz Centre for Ocean Research, Kiel, GEOMAR** Kiel, Germany
Computer Vision Research Assistant at Oceanic Machine Vision Group (OMV) Nov 2020 – June 2023
 - **Refractive Camera Models and Calibration:** Derive and implement novel camera models for underwater camera systems to account for geometric distortion induced by refraction (paper [5, 7]).
 - **Refractive Image-based 3D Reconstruction:** Research and develop a refractive structure-from-motion (SfM) approach for image-based 3D reconstruction from underwater imagery (paper [2]).
 - **Sensor Fusion with Navigation:** Develop a loosely-coupled visual-navigation fusion strategy for mapping a large-scale of seafloor (paper: [3]).
 - **Macro-lens Camera Calibration:** Derive a novel affine transform camera model for the focus-stacked image and develop a C++ based calibration approach for macro-lens camera system (paper: [6]).
- **Helmholtz Centre for Ocean Research, Kiel, GEOMAR** Kiel, Germany
Internship & Master Thesis Jan 2019 – Jan 2020
 - **BubbleBox Project:** Develop a synchronized high-speed (80 Hz) machine vision stereo camera system. Develop a C++ based software to measure and quantify the volumes of released gas bubbles using image processing and stereo vision techniques.
- **Computer Vision Freelancer** Kiel, Germany
Working as a freelancer to offer small-to-medium size computer vision services and solutions 2021 – 2024
 - **Localizing Robot Arm End Effector:** Develop a synchronized 4-camera system to track and localize the end effector of a robot arm in 3D using Aruco maker.
 - **Intrinsic & Extrinsic Calibration of a Camera Array:** Develop a calibration method to calibrate the intrinsic and relative extrinsic parameters of a 26-cameras camera array for medical human-body scanning.

SKILLS

- **Skill Set:** Camera Calibration; 3D Reconstruction; SfM / SLAM; Estimation and Optimization; Representation Learning;
- **Programming & Tools:** C++, CMake, Python, PyTorch, OpenCV, Ceres-Solver, Linux, Git, ROS (basic), LaTeX
- **Scientific Skills:** Problem Solving; Modeling; Scientific Drawing and Writing.
- **Communication:** English (proficient); German (A2); Chinese (native)

CERTIFICATE

- **Algorithmic Toolbox:** Issued by Coursera, credential ID: PX5W3GZW8QLW
- **Neural Networks and Deep Learning:** Issued by Coursera, credential ID: 6ML6ZBMNEXXU
- **Improving Deep Neural Networks:** Issued by Coursera, credential ID: JVRXWMCD4LHH

AWARDS & SCHOLARSHIP

- **Ph.D. Scholarship:** Doctoral scholarship granted by China Scholarship Council (CSC, 2020 – 2024)
- **Travel Grant:** Travel grant for young researchers by Deutsche Arbeitsgemeinschaft für Mustererkennung, DAGM 2019

SELECTED PUBLICATIONS

18 publications listed on Google Scholar, with a total of 199 citations and an H-index of 8. Below are selected publications:

- [1]: **M. She**, F. Seegräber, D. Nakath, P. Schöntag, K. Köser. Relative Illumination Fields: Learning Medium and Light Independent Underwater Scenes. (Submitted to *CVPR*, 2025)
- [2]: **M. She**, F. Seegräber, D. Nakath, K. Köser. Refractive COLMAP: Refractive Structure-from-Motion Revisited. In *IROS*, 2024 (**Oral**)
- [3]: **M. She**, Y. Song, D. Nakath, K. Köser. Semihierarchical Reconstruction and Weak-area Revisiting for Robotic Visual Seafloor Mapping. In *Journal of Field Robotics*
- [4]: **M. She**, T. Weiß, Y. Song, P. Urban, J. Greinert, K. Köser. Marine Bubble Flow Quantification Using Wide-baseline Stereo Photogrammetry. In *ISPRS Photogrammetry and Remote Sensing*
- [5]: **M. She**, D. Nakath, Y. Song, K. Köser. Refractive Geometry on Underwater Domes. In *ISPRS Photogrammetry and Remote Sensing*
- [6]: X. Weng*, **M. She***, D. Nakath, K. Köser (**Equal Contribution*). Macal - Macro Lens Calibration and the Focus Stack Camera Model. In *3DV*, 2021 (**Oral**)
- [7]: **M. She**, Y. Song, J. Mohrmann, K. Köser. Adjustment and Calibration of Dome Port Camera Systems for Underwater Vision. In *GCPR*, 2019 (**Oral**)
- [8]: D. Nakath, X. Weng, **M. She**, K. Köser. Visual Tomography: Physically Faithful Volumetric Models of Partially Translucent Objects. In *3DV*, 2024